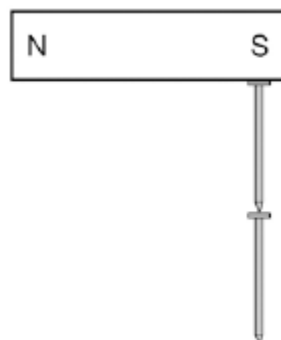


**Q1.**

**Figure 1** shows two iron nails hanging from a bar magnet.

The iron nails which were unmagnetised are now magnetised.

**Figure 1**



- (a) Complete the sentence.

Use a word from the box.

|               |                |                  |
|---------------|----------------|------------------|
| <b>forced</b> | <b>induced</b> | <b>permanent</b> |
|---------------|----------------|------------------|

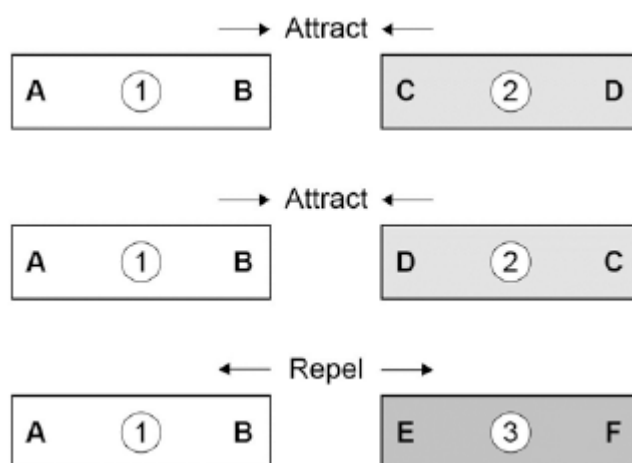
The iron nails have become \_\_\_\_\_ magnets.

(1)

- (b) Each of the three metal bars in **Figure 2** is either a bar magnet or a piece of unmagnetised iron.

The forces that act between the bars when different ends are placed close together are shown by the arrows.

**Figure 2**



Which **one** of the metal bars is a piece of unmagnetised iron?

Tick **one** box.

Bar 1



Bar 2



Bar 3



Give the reason for your answer.

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(2)

- (c) A student investigated the strength of different fridge magnets by putting small sheets of paper between each magnet and the fridge door.

The student measured the maximum number of sheets of paper that each magnet was able to hold in place.

Why was it important that each small sheet of paper had the same thickness?

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(1)

- (d) Before starting the investigation the student wrote the following hypothesis:

'The bigger the area of a fridge magnet the stronger the magnet will be.'

The student's results are given in the table below.

| Fridge magnet | Area of magnet in mm <sup>2</sup> | Number of sheets of paper held |
|---------------|-----------------------------------|--------------------------------|
| A             | 40                                | 20                             |
| B             | 110                               | 16                             |
| C             | 250                               | 6                              |
| D             | 340                               | 8                              |
| E             | 1350                              | 4                              |

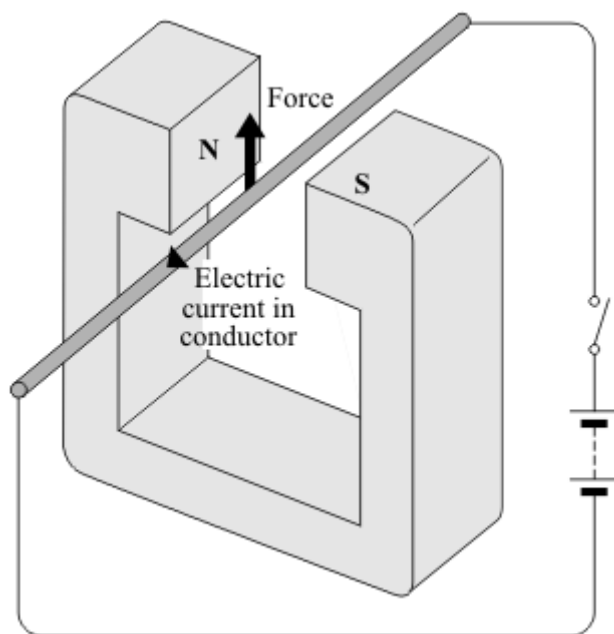
Give **one** reason why the results from the investigation **do not** support the student's hypothesis.

---

(1)  
(Total 5 marks)

**Q2.**

When a conductor carrying an electric current is placed in a magnetic field a force may act on it.



- (a) State **two** ways in which this force can be increased.

1. \_\_\_\_\_
2. \_\_\_\_\_

(2)

- (b) State **two** ways in which this force can be made to act in the opposite direction.

1. \_\_\_\_\_
2. \_\_\_\_\_

(2)

- (c) In what circumstance will **no** force act on a conductor carrying an electric current and in a magnetic field?

\_\_\_\_\_  
\_\_\_\_\_

(1)

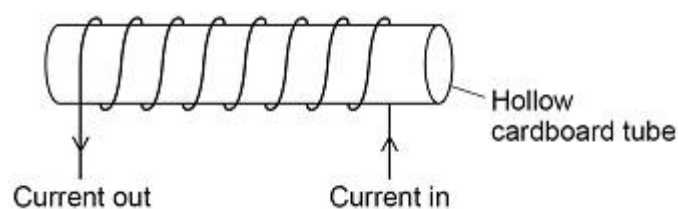
(Total 5 marks)

**Q3.**

- (a) **Figure 1** shows a solenoid.

Draw the magnetic field of the solenoid on **Figure 1**.

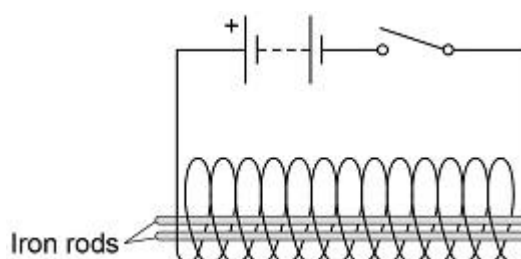
**Figure 1**



(2)

(b) **Figure 2** shows two iron rods placed inside a solenoid.

**Figure 2**



Explain why the iron rods move apart when the switch is closed.

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(2)

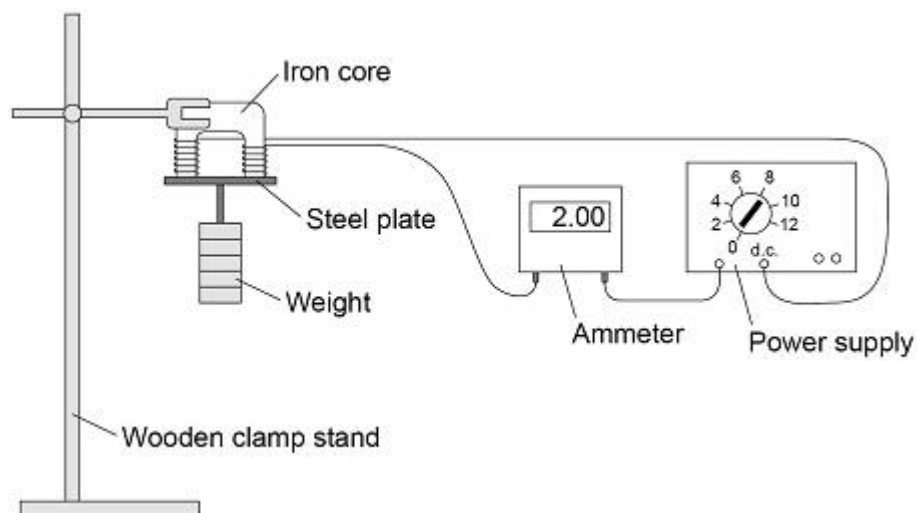
A student investigated the strength of an electromagnet.

The student investigated how the strength depended on:

- the current in the wire
- the number of turns of wire around the iron core.

**Figure 3** shows the equipment used.

**Figure 3**



The student measured the strength of the electromagnet as the maximum weight the electromagnet could hold.

(c) The following table shows the results.

| Current in<br>amps | Number of turns of<br>wire | Maximum weight<br>in newtons |
|--------------------|----------------------------|------------------------------|
| 1.0                | 30                         | 6.5                          |
| 1.5                | 20                         | 6.4                          |
| 2.0                | 10                         | 3.7                          |

Explain why the method used by the student is **not** valid for this investigation.

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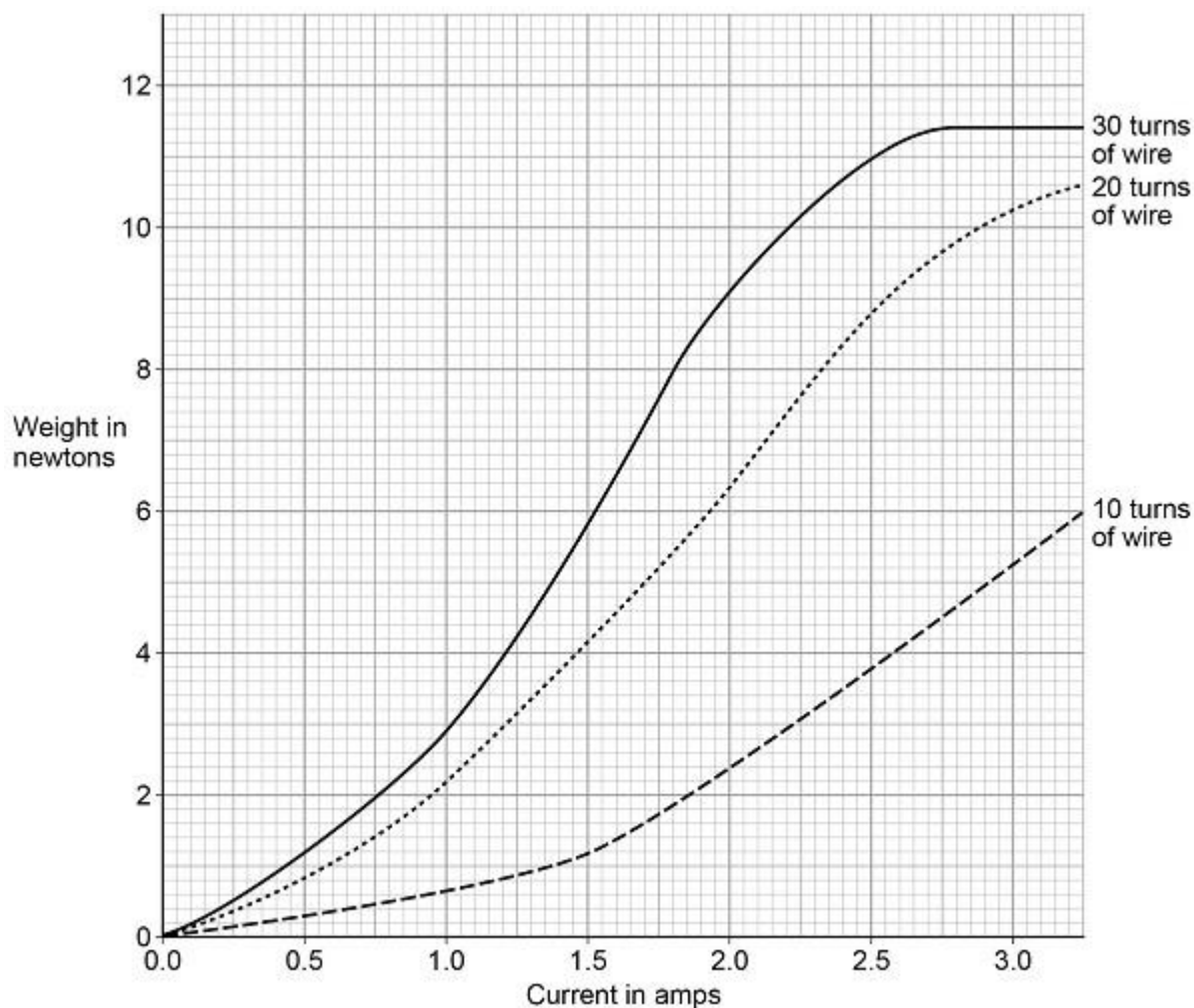
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(2)

A second student repeated the investigation using the same equipment.

**Figure 4** shows the second student's results.

**Figure 4**



- (d) How does increasing the current in the wire affect the strength of the electromagnet, when the electromagnet has 30 turns of wire?

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(1)

- (e) How does increasing the number of turns of wire from 10 to 20 affect the strength of the electromagnet, compared to increasing the number of turns of wire from 20 to 30?

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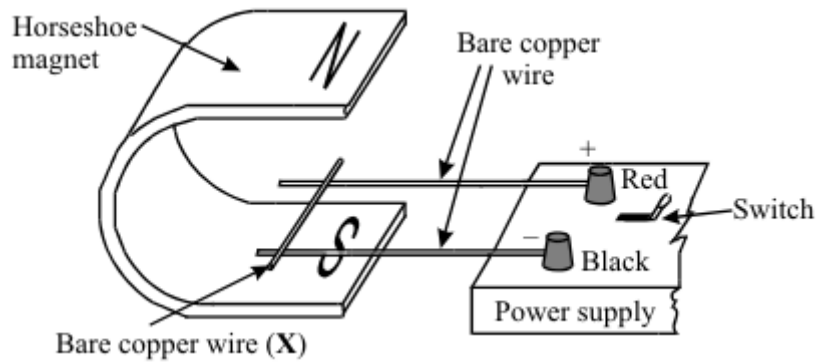
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(1)

(Total 8 marks)

**Q4.**

The diagram shows apparatus used to demonstrate the motor effect. **X** is a short length of bare copper wire resting on two other wires.



- (a) (i) Describe what happens to wire **X** when the current is switched on.

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- (ii) What difference do you notice if the following changes are made?

A The magnetic field is reversed.

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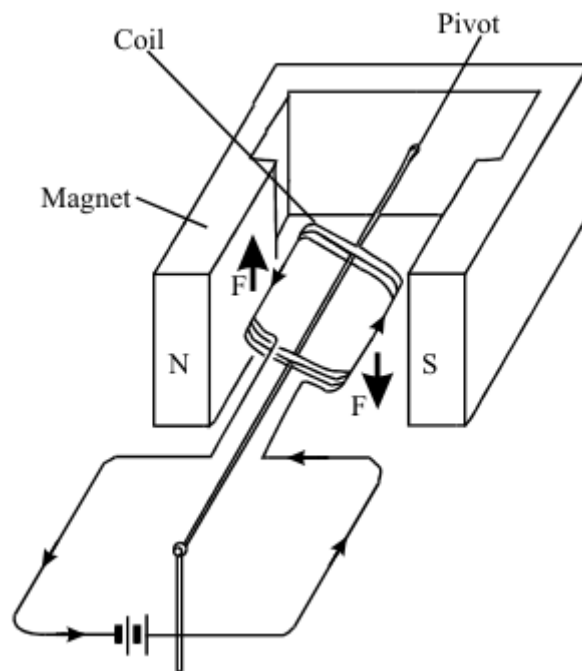
B The current is increased.

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(3)

- (b) The diagram shows a coil placed between the poles of a magnet. The arrows on the sides of the coil itself show the direction of the conventional current.



The arrows labelled **F** show the direction of the forces acting on the sides of the coil. Describe the motion of the coil until it comes to rest.

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(3)

- (c) Most electric motors use electromagnets instead of permanent magnets. State three of the features of an electromagnet which control the strength of the magnetic field obtained.

1. 

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2. 

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3. 

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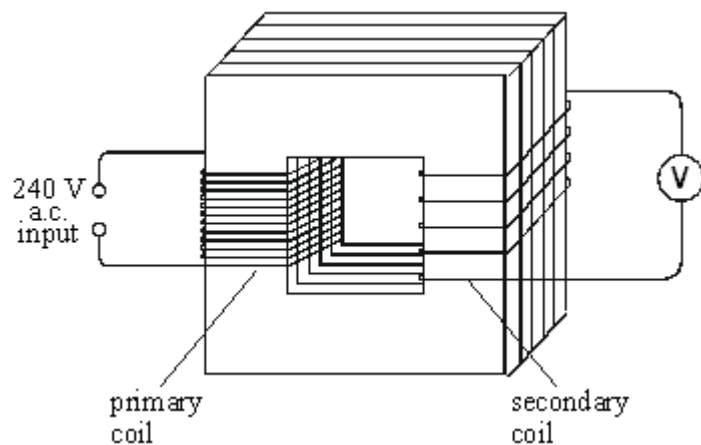
(3)

(Total 9 marks)

### Q5.

The diagram below shows a transformer.





- (i) Name the material used to make the core of the transformer.

\_\_\_\_\_

(1)

- (ii) The primary coil has 48 000 turns and the secondary coil 4000 turns.

If the input voltage is 240 V a.c., calculate the output voltage.

\_\_\_\_\_

\_\_\_\_\_

Answer \_\_\_\_\_ V

(2)

- (iii) Explain how the use of such a transformer could be adapted to transform a low voltage into a higher voltage.

\_\_\_\_\_

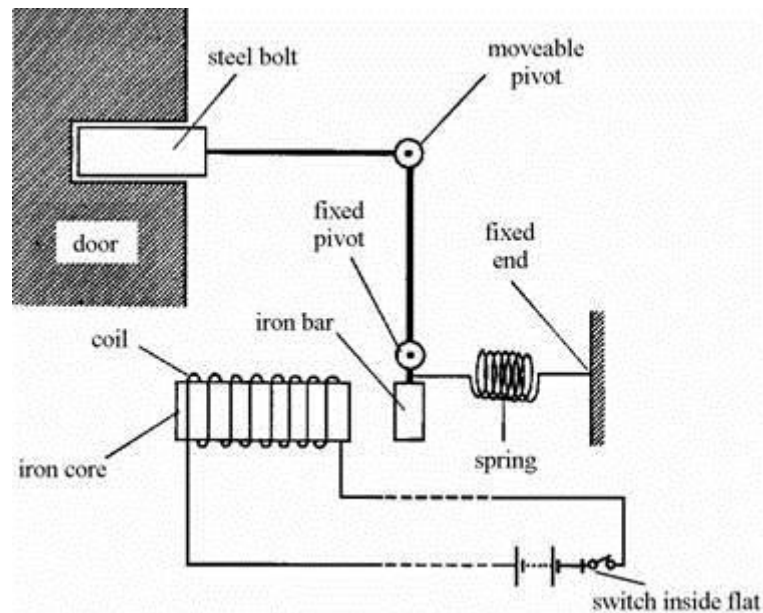
\_\_\_\_\_

(1)

(Total 4 marks)

## Q6.

The diagram below shows a door lock which can be opened from a flat inside a building.



- (a) Explain how the door is unlocked when the switch is closed.

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(4)

- (b) State **two** changes which would increase the strength of the electromagnet.

1. 

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2. 

---

(2)

- (c) Why is the spring needed in the lock?

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(1)

- (d) The connections to the coil were accidentally reversed. Would the lock still work?

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Explain your answer.

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(2)

(Total 9 marks)

## Mark schemes

### Q1.

(a) induced 1

(b) bar 2 1

(the same end) of bar 1 attracts both ends of bar 2

**or**

only two magnets can repel so cannot be bar 1 or bar 3 1

(c) so the results for each magnet can be compared

**or**

so there is only one independent variable

*fair test is insufficient*

*allow different thickness of paper would affect number of sheets each magnet could hold*

*accept it is a control variable*

1

(d) because the magnet with the biggest area was not the strongest  
*accept any correct reason that confirms the hypothesis is wrong eg smallest magnet holds more sheets than the largest*

1

[5]

### Q2.

(a) increase the current (1)  
*credit increase the p.d./voltage*  
*credit reduce the resistance*  
*credit have thicker wiring*  
*credit add extra / more cells* 1

increase the magnetic field (strength) (1)  
*credit 'have stronger magnet(s)'*  
*do not credit 'bigger magnets' either order*

1

(b) **either** reverse polarity

**or** connect the battery the other way round 1

**either** reverse direction of the magnetic field

**or** put the magnet the other way round / reverse the magnet

*do **not** give any credit to a response in which both are done at the same time either order*

1

(c) **either**

conductor parallel to the magnetic field

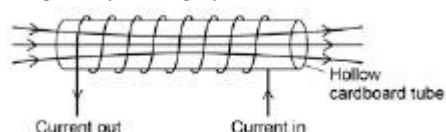
**or** lines of magnetic force and path of electricity do not cross

1

[5]

### Q3.

(a) field lines going in, (through) and out of the solenoid



*allow field lines only visible outside the cardboard tube*

*allow a bar magnet shaped field with lines above and below the solenoid*

1

arrow(s) in correct direction

1

(b) the rods become (induced) magnets

*allow the rods are (temporarily) magnetised*

*ignore rods repel*

*do **not** accept rods become charged*

1

with the same polarity (at each end)

1

(c) changed two (independent) variables (at the same time)

*allow need to keep current or number of turns constant*

*allow should only change one variable (at a time)*

*allow current and number of turns both changed*

*ignore fair test*

1

so it is not possible to know the effect of one (independent) variable or the other

1

(d) (increasing the current) increases the strength until the strength reaches a maximum value

*allow weight (held) for strength of electromagnet*

*ignore a given current value for when maximum strength happens*

1

- (e) increasing the number of turns from 10 to 20 increases the strength more than increasing from 20 to 30

*a general trend is required*

1

[8]

#### Q4.

- (a) (i) it moves or experiences a force horizontally to the right  
*for 1 mark*

1

- (ii) A – moves in opposite direction or force reversed e.c.f.  
B – faster movement or larger force  
(**not** move further)  
*for 1 mark each*

2

- (b) turns clockwise  
oscillates/reverses  
comes to rest facing field/at 90° to field/vertically  
*for 1 mark each*

3

- (c) number of turns or linear number density of turns current core  
*for 1 mark each*

3

[9]

#### Q5.

- (i) iron  
*for 1 mark*

1

- (ii) 20  
*gains 2 marks*

else working  
*gains 1 mark*

2

- (iii) reverse input/output  
*for 1 mark*

**or** increase secondary turns

1

[4]

#### Q6.

- (a) current flows  
coil / core magnetised / electromagnet activated / energised / turned on  
attracts iron bar causing bolt to be pulled out

*each for 1 mark*

4

- (b) more turns  
bigger current / e.m.f  
softer iron core

*any two for 1 mark each*

2

- (c) to relock door / return iron bar / to lock door  
*for 1 mark*

1

- (d) iron bar would still be attracted / coil still magnetised so still works  
*for 1 mark each*

yes + wrong answer  
*0 marks*

yes + current still flows  
*1 mark*

yes + still magnetised / iron bar still attracted  
*2 marks*

2

[9]

## Examiner reports

### Q2.

- (a) Most candidates chose 'increase the current' or a correct way of doing this, such as to increase the p.d. or increase the number of cells, and some also suggested 'have a stronger magnet'. However writing 'have a bigger magnet' did not gain credit as a bigger magnet would not necessarily be stronger and the poles could be further from the conductor.
- (b) Some candidates correctly suggested that the battery would need to be connected the other way round or that the magnet would need to be put the other way round. However some answers were so poorly expressed that it was impossible to be reasonably sure what was being suggested.
- (c) Few candidates knew that there would be no force on the conductor when the conductor is parallel to the magnetic field or, put another way, when the lines of magnetic force and the path of the electricity do not cross.

### Q3.

- (a) Just over 11% of the students did not attempt this question. The majority of field patterns drawn showed a lack of precision. Lines joining and crossing at the ends were common. Most patterns that scored the first mark point were a bar magnet type field pattern. Confusion of arrow direction was very common with arrows going into and out of an end. Some students did not appear to understand the term 'solenoid'.
- (b) This question was not well understood. About half of the students understood that the rods became induced magnets but often then just wrote that they repelled with no mention of the idea of same polarity/like poles. Very few answers were really clear about like poles being on adjacent ends. Incorrect ideas that were common included that the rods were charged, with phrases such as 'charged with magnetism' used, as was the idea that they were attracted to the sides of the solenoid or were repelled apart when the current was switched off. Some students stated that a current was in the rods or that closing the switch meant that current was stopped and the solenoid was no longer magnetised.
- (c) Many of the students did recognise that two variables had been changed at the same time.

Only around 18% of the students scored the second mark.

- (d) About half of the students scored this mark. Those that did not score often failed to mention that the strength reached a maximum value.
- (e) Nearly three quarters of the students scored this mark, recognising that the increase in strength from 10-20 turns was higher than the increase from 20-30. Some that did not score the mark simply quoted numbers from the graph but did not use them in any comparative way to show a general trend as they were for a specific point. Others that did not score the mark usually gave a description of the shapes and/or gradients of the three graphs with no comparison of the increase in strength.

### Q4.

In part (a)(i) very many candidates had the wire moving, unfortunately in many cases the

movement was up or down. Few candidates stated that the wire would move along the copper wires to the right. In part (ii) many had the movement reversed, but reversing an up/down movement does not give a correct answer only answers stating that the wire would move to the left were acceptable. In the last part of the question a faster movement was the common correct answer. In part (b) candidates seemed to be familiar with the electric motor but experienced difficulty in applying their ideas in an unfamiliar situation. Most candidates started their answer with a clockwise movement for one mark, fewer went on to say the coil would then vibrate backwards and forwards for the second mark. The final mark, for stating (in their own words) that the coil would end up perpendicular to the field, was gained by a small number of candidates. Part (c) was quite well answered with most candidates scoring marks for increasing the number (or number density) of turns and the current. many went on to gain the third mark for the use of a soft iron core.

#### **Q5.**

The answers to this question were generally pleasing. In (iii) the expected scoring response was to reverse the input and output terminals but full credit was also given to the many candidates who correctly rewired the transformer.